RAHEE

Week 4 Project Progress Report (Final Report)

Project	RAHEE, an online digital version of a travel board game
Team	Team 31 (5 members)
Report period	Week 4 (9 October - 16 October 2026)

Current stage

This is the final report. All three bugs raised at the 7 October meeting are fixed and are now covered by automated checks. Looking into the Gwalior bug led us to audit every road on the board against the printed map, and that found and corrected the same kind of fault on several other routes. The panel showing the other players' destinations and reconnection after a dropped connection are both built, so every item on the Week 3 plan is now complete. We re-ran the full test suite and played complete games on the final build, from the browser and from scripted clients, with 2, 3, 4 and 6 players. The game is split into a static frontend, a Node backend and a MySQL-compatible database, and has been deployed following the steps in the README.

1. Work Completed This Week

All of the changes below were made and tested properly. Items 1 to 4 fix the bugs from the 7 October meeting, item 5 adds the tests that keep them fixed, and items 6 to 9 close the new requirements and the known issues listed in Week 3.

1. **Flights are now a last resort (bug B2):** The route search first looks for a road or ship route and only uses a flight when no such route fits inside the played card's allowance. Which cities can be reached has not changed, only which route is shown. New Delhi to Mumbai is now the six-step road route through Agra, Gwalior, Bhopal, Aurangabad and Daman instead of a single flight. If the card allows only one step, the flight is still offered because no road route fits.
2. **Spare 10-valued cards are set aside (bug B3):** After every player has received their guaranteed 10 (or J in the 5 and 6 player games), all the other 10s and Js are removed from the deck instead of being shuffled back in. Each hand now holds exactly one high-movement card. The deal also returns the set-aside pile so a test can check it.
3. **Route fix around Gwalior (bug B1):** The board data had a Bhopal-Lucknow road that does not exist on the printed board. The printed map has Bhopal-Gwalior and Gwalior-Lucknow. That single wrong road let the search skip Gwalior. We removed it, and Sawai Madhopur to Madikeri in four steps now goes through Gwalior (Sawai Madhopur, Gwalior, Lucknow, flight to Bengaluru, Madikeri).
4. **Board-wide route audit and corrections (new bug B4):** The same fault was present in other places. The tool that traced the roads off the artwork had followed two printed roads end to end through a city and recorded them as one long road, so a route between the two outer cities skipped the city in the middle. We checked every road on the board against the printed

map and corrected all of them. The eight roads removed are listed in the table below, each now reached through the city in between.

Road removed	Now goes through	Why
Bhopal - Lucknow	Gwalior	Bhopal-Gwalior and Gwalior-Lucknow were recorded as one road
Kochi - Madikeri	Wayanad	Kochi-Wayanad and Wayanad-Madikeri
Badrinath - Rishikesh	New Delhi	Badrinath-New Delhi and New Delhi-Rishikesh
Bengaluru - Wayanad	Udupi (or Madikeri)	Bengaluru-Udupi and Udupi-Wayanad
Daman - Pune	Mumbai	Daman-Mumbai and Mumbai-Pune
Madurai - Thiruvananthapuram	Kanniyakumari	No printed road between the two
Dharamshala - Jammu	Amritsar	No printed road between the two
Coimbatore - Wayanad	Kochi	Wayanad-Kochi and Kochi-Coimbatore

We also checked the opposite case. **Amritsar-Jammu** and **Aizwal-Imphal** pass close to Dharamshala and Silchar, but the printed map shows each as its own road, so they were kept. Two roads that were missing were added (Nagapattinam-Madurai and Kochi-Coimbatore), and the stored drawn path of ten roads was redrawn to follow the printed road (Aurangabad-Bhopal, Lucknow-Gorakhpur, Gulbarga-Ratnagiri, Siliguri-Guwahati, Gandhinagar-Vadodara, Bilaspur-Rourkela, Srinagar-Jammu, Udupi-Wayanad, Hisar-Jaipur and Hisar-New Delhi). The board now has 204 routes: 153 roads, 48 flights and 3 ships. A final check of all 153 roads found none that passes within 30 map pixels of a city it does not stop at, apart from Nagapattinam-Madurai, which runs close to Thanjavur but is a separate printed road. Every city can be reached, and 96 of the 101 can be reached by road alone. The other five are the island cities, which the printed board joins only by flight or ship.

5. **Regression tests for B1 to B4:** The test script (tools/test-engine.mjs) now checks that no spare 10 or J is dealt or left in the deck, for every player count from 2 to 6 and across 200 different shuffles each. It also checks that New Delhi to Mumbai is a road route when the card allows it and a flight only when the card allows one step, that every removed road stays removed and is replaced by its two-road link, that the kept roads are still present, and that Sawai Madhopur to Madikeri passes through Gwalior. The script now runs 3,514 checks in total.
6. **Other players' destinations panel (Week 3 requirement 1):** A tab on the left edge of the map opens a panel called "Where everyone is heading". It has one row for each other player showing their colour, name, character, current city and how many of their six destinations they have reached. A row opens to show the six destinations with their short descriptions and tourist numbers, with a tick on the ones reached. Tapping a place moves the map to it. The panel opens on the player whose turn it is, and rows are tagged "playing" or "offline". It uses only public game state, so nothing extra is requested from the server and hands stay hidden.
7. **Reconnection to a seat (Week 2 and Week 3 known issue):** A player whose connection drops can now return to the same seat and carry on. Each seat gets a secret token when the player

creates or joins a room, and the browser keeps it for the tab. The server stores only a hash of the token in the database. A reload or a dropped connection reconnects by itself with a growing delay (half a second up to eight seconds), and straight away when the network returns or the tab is shown again, under a "Connection lost - reconnecting" banner. A player who closed the tab enters the room code and the same name to get the seat back with a fresh token and only a seat with no live connection can be taken back this way. If the same seat is opened in a second window, the newest window wins and the older one is told so and offered a button to take the seat back. Everyone sees an "offline" tag on a player who is away, and the waiting screen says so when the game is waiting on them. A server heartbeat every 25 seconds drops dead connections so that a seat is not shown as online forever. Because seats and tokens are saved in the database, this also works after a server restart.

8. **Map information and artwork in the game (Week 3 requirement 2):** The game shows the printed board with every city as a node, the six printed character cards in the lobby and during play, and a hover panel on every city giving its tourist number, place name and attraction, an airport marker, the cost of the next step, whether it is your destination, and who is standing there.
9. **Undo of a click-to-route walk:** Clicking the destination again while still standing on it now undoes the whole walk in one go, instead of one step at a time.

Status of the items planned in Week 3

#	Item (from the Week 3 plan)	Status
1	Make the route search prefer road paths and offer a flight only when no road path fits	Completed (item 1 above)
2	Set aside the leftover 10-valued cards and test it for every player count	Completed (item 2 above)
3	Re-check the Bhopal-Lucknow link and the routes around Gwalior against the printed board	Completed, extended to the whole board (items 3 and 4)
4	Add regression tests for B1, B2 and B3	Completed (item 5 above)
5	Click-to-open panel showing other players' destinations	Completed (item 6 above)
6	Add the new artwork and node map to the game	Completed (item 8 above)
7	Reconnection so a dropped player can rejoin their seat	Completed (item 7 above)

Deployment is complete: RAHEE is deployed on Render and can be accessed online. The project configuration keeps database settings in environment variables and the backend server address in one place. The README documents the relevant configuration and deployment steps. No additional hosting platform or separate frontend deployment is required for the current setup.

2. Bugs Identified

The three bugs below were raised at the 7 October meeting. B4 was found by us while fixing B1. All four are fixed.

ID	Bug	What we changed	Status	Checked by
B1	The route from Sawai Madhopur to Madikeri did not pass through the Gwalior node	Removed the Bhopal-Lucknow road, which is not on the printed board, so the Bhopal-Gwalior and Gwalior-Lucknow roads are used	Fixed	Test in the test script; route drawn over the map
B2	A flight was offered as the shortest path even when a road path existed	Search tries roads and ships first and falls back to a flight only when nothing else fits the card	Fixed	Test in the test script; New Delhi to Mumbai
B3	After the guaranteed 10-valued cards were handed out, the leftover 10-valued cards were shuffled back into the deck	Spare 10s and Js are now set aside before the rest of the deck is dealt	Fixed	Test for 2-6 players, 200 shuffles each
B4	Eight roads skipped the city they pass through, so routes between the outer cities did not stop there	Removed the eight roads, added two missing roads and redrew ten road paths	Fixed	Tests for each road; audit of all 153 roads

Known issues carried over from Week 3

Issue	Status
A player whose connection drops cannot rejoin their seat	Closed (item 7)
Every route uses a step value of one	Closed. The rulebook defines one step as one route segment between two adjacent cities, and every road, flight and ship on the board is one segment
Undo only reverses the last single movement step	Improved. A whole click-to-route walk can be undone at once. A finished move or a whole turn still cannot be undone and not actually needed

3. Requirements from the Meeting

Both new requirements from 7 October are done.

Requirement	Status
Destinations of other players visible in a click-to-open panel on the left	Completed. The panel shows the six destinations on each character card
New artwork: node map, character cards and the board for the left panel	Completed. The board with its nodes and the six character cards are in the game, with hover information on every city

4. Future Work

The Week 3 plan is complete and RAHEE is deployed on Render. Remaining work is limited to optional extensions and any issues found during further validation:

- Validate the deployed application across browsers and networks, including multiplayer sessions, reconnection, and persistence with the configured production database.
- Optional extensions include a visual map editor, a computer opponent, route hints, movement animation and play-test statistics.
- Other possible improvements are whole-turn undo and updating the browser walkthrough tools for the current lobby flow.

5. Time Spent by Each Team Member, last Week

All of the Week 4 changes were made by one team member, so the table lists only that member.

Name	Task(s)	Approx. Hours/week
Tanuj Sharma	Route-search and dealing-rule fixes; review of movement behaviour and regression cases (engine.js, test-engine.mjs).	3
Sai Gawali	Room/session and reconnection review; verification of seat recovery, connection status and multiplayer flow (rooms.js, handlers.js, protocol.js).	3
Siddhesh Patil	Backend, persistence and multiplayer testing; review of game-state restoration and server-side rule enforcement (db.js, app.js, handlers.js).	3
Mukesh Dewangan	Frontend integration and interface review, including the destinations panel, board artwork and deployment configuration (main.js, sidebar.js, config.js).	3
Kshitij Kasodkar	Board-wide route audit and corrections, regression-test review, final testing and report consolidation (build-board.mjs, route-fixes.json, board.json, test-engine.mjs).	3

6. Last Meeting with the POC

Date of last meeting	7 October 2026 (Wednesday), 2:00 pm
Meeting cadence	Weekly, every Wednesday at 2:00 pm.
Mode	Offline

Key discussion / action items

Key discussion / action items from the 7 October meeting:
1. Correct the Sawai Madhopur–Madikeri route so it passes through Gwalior; completed.

2. Add a click-to-open panel showing other players' destinations; completed.
3. Set aside leftover 10-valued cards after the guaranteed cards are dealt; completed.
4. Prefer road/ship routes and offer a flight only when no suitable road/ship route fits the card; completed.
5. Integrate the supplied board/node artwork and character cards; completed.

7. Date of the Current Build

Current build date	15 October 2026
Build / version identifier	1a62380, main branch

8. Test Results on the Current Build

Tests are run against the rules engine with a script that plays a large number of complete random games and checks the rules after every one of them. The script was re-run on the current build.

Total tests	80 complete random games, each checked against every rule (deck composition, legal movement, scoring, and the two worked examples from the manual)
Unsuccessful tests	0
Successful tests	80
Other checks in the same script	3,514 checks in total, including the regression checks for B1 to B4. 0 failed

The following playthroughs were also run on the final build:

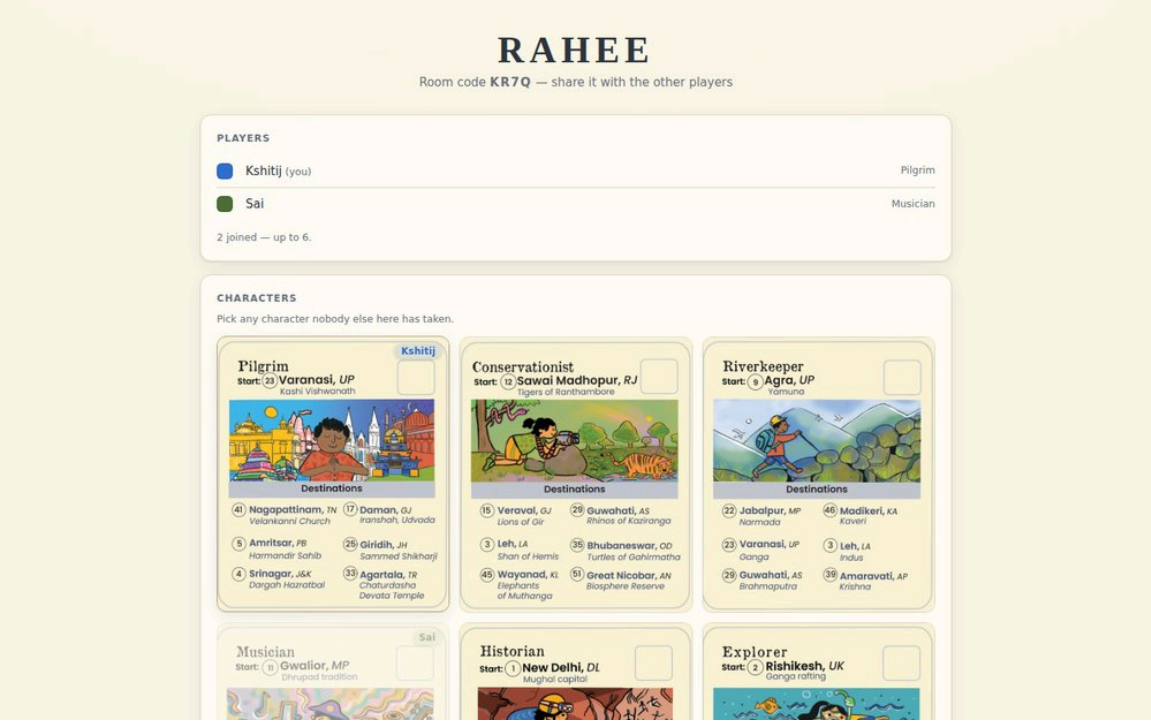
Test	Result
Full games through the server, 2, 3, 4 and 6 players	All four played to the final scoreboard. Road trips with accept, counter and decline replies, rider-chosen drop-offs, shared tourism and Aces were all used. Passed
Privacy	The full board, the undealt deck and the shuffle seed were never sent. Every other player's hand arrived only as a card count, for every player in every game. Passed
Turn control	An action sent by a player whose turn was, it was not rejected with "it isn't your turn". Passed
Reconnection	A dropped player resumed with the token and had an identical hand. A wrong token and an unknown room got the same answer. A second window took over the seat and the old window was closed. Room code plus name gave the seat back with a new token and the old token stopped working. A connected seat could not be taken. Passed
Server restart in mid-game	The room was reloaded from the database with identical game state, all players resumed, and the game was played to the end. Passed

Test	Result
Browser playthrough	Three browsers played a 3-player game to the Journey's End scoreboard, and one page was reloaded mid-game and returned to its seat. Two browsers also went through the lobby, laying the first card, clicking a destination, the destinations panel, and a road trip invitation with a rider-chosen stop. No browser errors. Passed

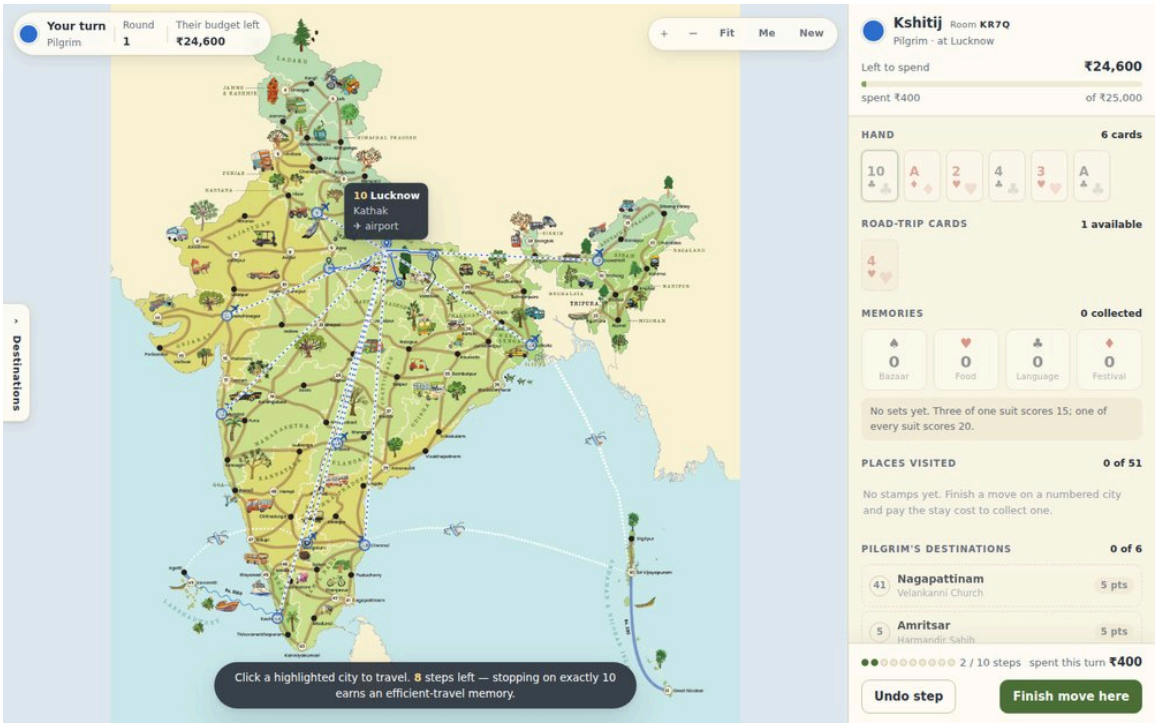
These playthroughs ran on the real backend code with an in-memory stand-in for the database and the WebSocket library, so the MySQL or TiDB connection itself is only used when the backend is started with real database settings. The random-game script checks the rules but not which route the search picks, so route choice is covered by the B1 to B4 checks in the same file. The browser and reconnection playthroughs were run as scripts for this report and are not part of the project folder.

Screenshots of the game

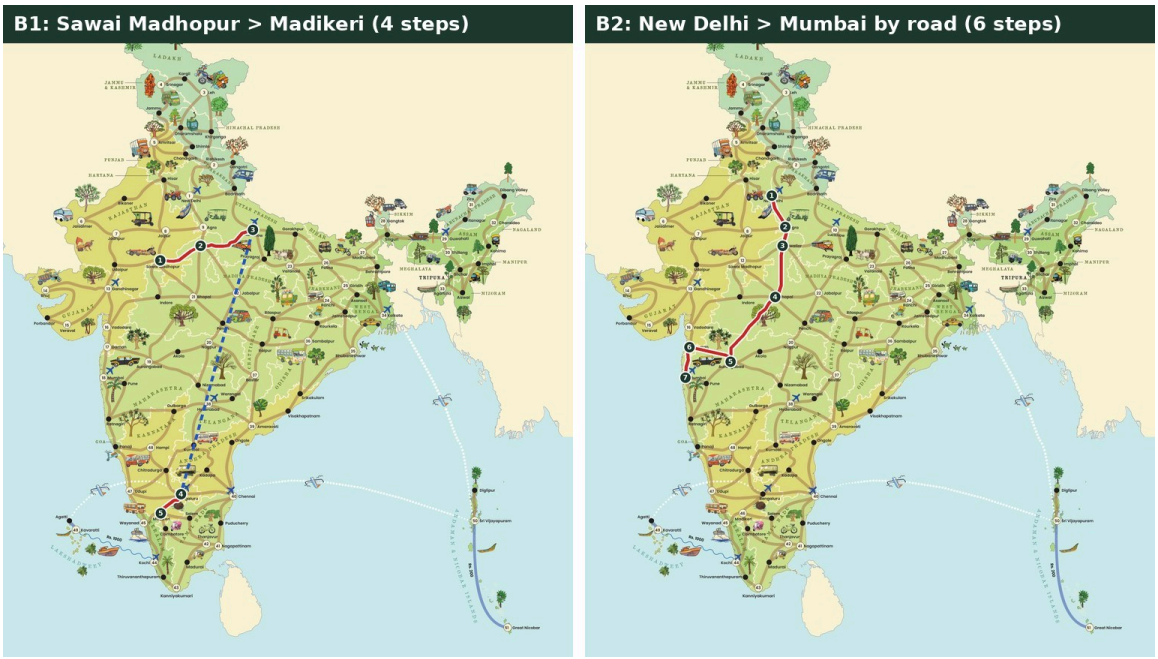
1. Room lobby with the printed character cards (host view)



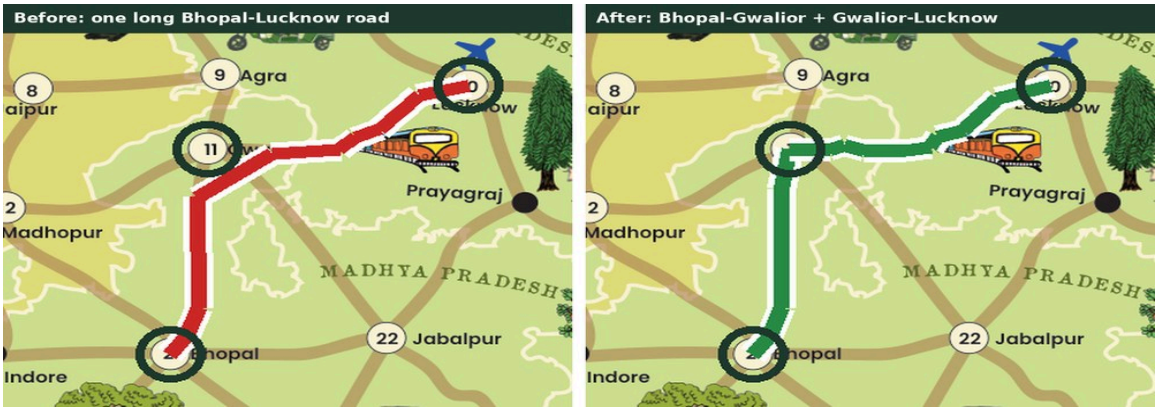
2. Destination clicked on the map: the shortest route is walked (Varanasi to Lucknow, 2 steps, spent so far shown)



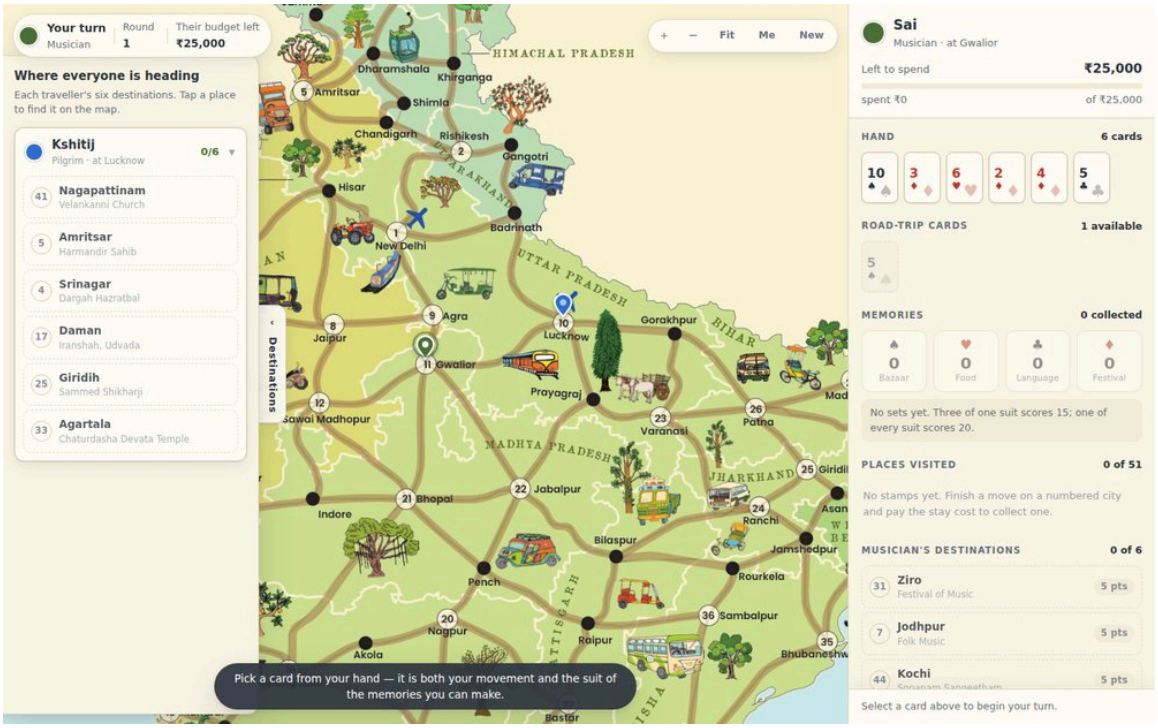
3. Bug fixes B1 and B2, drawn from the board data: Sawai Madhopur to Madikeri through Gwalior (left) and New Delhi to Mumbai by road instead of a flight (right)



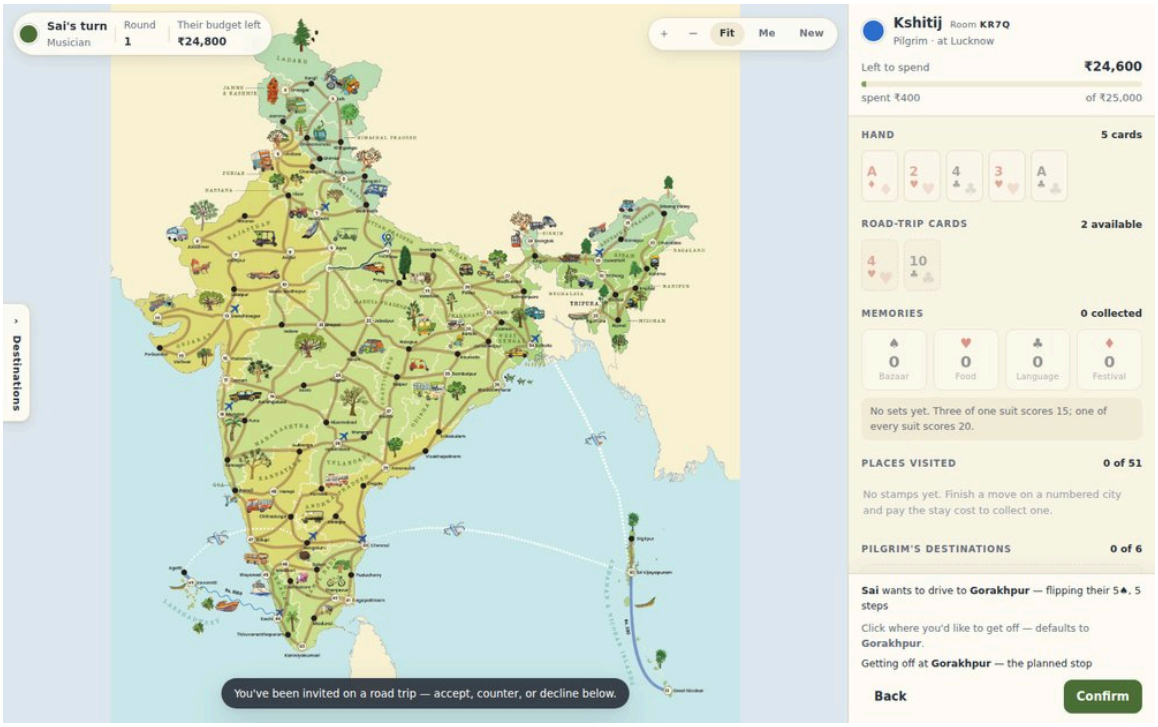
4. Bug fix B4: the single long Bhopal-Lucknow road that let routes jump past Gwalior (left) and the two real roads that replace it (right)



5. Other players' destinations panel opened from the left edge



6. Invited player choosing where to get off the road trip



7. Final scoreboard, Journey's End

